

CHEN FENG FAN

AI Solutions Developer | Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Results-driven Full-Stack Software & AI Engineer with enterprise experience developing scalable web, mobile, and AI-powered applications. Proficient in transforming complex business requirements into robust end-to-end solutions using Laravel 11, React, React Native, Flutter, and Node.js. Specialized in NLP/RNN-based intent analysis, Retrieval-Augmented Generation (RAG) pipelines, vector database architecture, graph pathfinding optimization, and legacy system migration. Passionate about building user-centric software that drives operational efficiency and delivers practical business outcomes.

TECHNICAL EXPERTISE

- **Artificial Intelligence & Machine Learning:** Recurrent Neural Networks (RNN/LSTM), Intent Analysis & Classification, Word & Text Detection, LangChain, Pinecone (Vector Database), Retrieval-Augmented Generation (RAG), Semantic Search, Prompt Engineering, Scikit-learn, Hugging Face, IBM watsonx.ai, LLM API Integration (OpenAI, Claude, Gemini)
- **Algorithms & Optimization:** Graph Data Structures & Modeling, Pathfinding Optimization (Dijkstra, A* Search), Relational Query Logic, Database Locking & Concurrency Control, SQL Aggregation
- **Programming Languages:** Python, PHP, JavaScript, TypeScript, SQL, C#, Rust, HTML5, CSS3
- **Frontend & Mobile Development:** React, React Native, Flutter, Tailwind CSS, Bootstrap, Responsive Web Design
- **Backend & Infrastructure:** Laravel 11, Node.js, Express.js, RESTful API Development, Docker, Git, Linux, JWT, Google OAuth
- **Web Scraping & Automation:** Playwright, Browser Automation, Web Data Extraction, Cron Job Scheduling, Automated Data Collection Pipelines
- **Browser Extension & AI Integration:** Chrome Extensions (Manifest V3), JavaScript, REST APIs, LLM Integration, Embeddings, RAG, LanguageTool
- **Database Systems:** PostgreSQL, MySQL, SQL Server, Database Design, Query Optimization, Schema Migration, Eloquent ORM
- **Methodologies & Tools:** Business Analysis, Agile Project Management, User Acceptance Testing (UAT), UX Design, Role-Based Access Control (RBAC), Object-Relational Mapping (ORM)

FEATURED AI & TECHNICAL PROJECTS

Enterprise RAG & Intent Classification AI Assistant

Lead AI Developer | Python, Recurrent Neural Networks (RNN), LangChain, Pinecone, Semantic Search, Hybrid Retrieval

- Architected an enterprise AI knowledge assistant featuring an Intent Analysis engine using Recurrent Neural Networks (RNN) to process user word sequences, detect query intent, and route requests dynamically.
- Engineered end-to-end LangChain pipelines and designed vector database architecture integrated with Pinecone for real-time semantic document retrieval across complex enterprise data.
- Optimized retrieval precision and answer accuracy significantly by engineering a hybrid search mechanism combining vector embeddings with structural SQL relational query filters.

Optimized Transit Network & Pathfinding Engine

Lead Software Developer | Python, Graph Data Structures, Pathfinding Algorithms (Dijkstra/A), Node & Edge Modeling*

- Designed and implemented an optimized network graph architecture modeling complex transportation nodes and edge connections.
- Integrated optimized pathfinding algorithms (Dijkstra / A*) to compute shortest paths, minimize transfer delays, and maximize overall routing efficiency.

Lion Befrienders Volunteer Management Platform

Full-Stack Developer | Node.js, Express.js, PostgreSQL, Google OAuth, REST APIs, RBAC

- Conducted deep user requirement analysis and formulated relational database architecture to design and build a comprehensive volunteer management ecosystem consisting of three independent portals (Volunteer, Institution, Admin).
- Implemented granular Role-Based Access Control (RBAC), robust messaging channels, registration workflows, event management modules, and secure Google OAuth authentication.

PROFESSIONAL EXPERIENCE

Silverback Aircon — Software & Automation Developer

Singapore | 2026 – Present

Web Scraping & Browser Automation

- Developed automated web scraping and browser automation systems using Playwright to extract and process business data from online platforms.
- Built Cron-scheduled workflows that independently execute platform operations and collect new information daily without manual intervention.

AI-Powered WhatsApp Customer Support Chrome Extension

- Built an internal WhatsApp Web Chrome extension that converts historical company conversations into embeddings and retrieves relevant context and company rules for RAG-based, context-aware LLM reply suggestions.
- Integrated LLM-assisted grammar correction with LanguageTool and developed Quick Reply workflows to improve response speed, grammar, clarity, and consistency.
- Published the extension through the Chrome Web Store for internal company use.

CBM Pte Ltd — Application Developer Intern

Singapore | March 2026 – Present

Building Inspection & Patrol Management System (CBM Qatar)

- Migrated legacy building patrol modules from Voyager to a modern React, Laravel 11, and Flutter architecture to support enterprise facility management operations.
- Designed and developed secure RESTful APIs implementing CRUD operations, Eloquent ORM, and Role-Based Access Control (RBAC).
- Integrated QR-code-based patrol workflows and built an administrative dashboard for real-time patrol monitoring.
- Optimized backend queries, input validation, and pagination to improve system performance, data security, and maintainability.

Parking Container Management & Finance System

- Modernized the container parking platform by migrating the legacy Voyager system to Laravel 11 and React.
- Developed core finance and invoice management modules, including an interactive revenue reporting dashboard driven by advanced SQL aggregation and database query optimization.
- Enhanced user experience and operational workflow by implementing optimized mobile image processing, interactive image zoom features, and fast search retention mechanisms.

Project Management & Reporting System

- Migrated legacy Voyager reporting platforms to secure progress tracking for daily and monthly construction and facility operations.
- Designed robust reporting workflows, file uploads, image management, and advanced multi-level form validation mechanisms.
- Implemented database locking, transaction management, and concurrency control to eliminate duplicate report submissions and guarantee financial/reporting data integrity.
- Integrated web and mobile applications with backend architecture, performing comprehensive User Acceptance Testing (UAT) with cross-functional project teams.

EDUCATION

Ngee Ann Polytechnic | Singapore | 2024 – Present
Diploma in Information Technology (AI Specialization)
Diploma Plus: Advanced Computing Mathematics

CERTIFICATIONS

Google Professional Certifications: Business Analysis & Process Management | Agile Project Management | Foundations of Project Management | Foundations of User Experience (UX) Design

LEADERSHIP & COMMUNITY INVOLVEMENT

Jurong SDA Church — Group Leader | Singapore | 2023 – Present